

Certifiably Correct Generalized Robot-World and Hand-Eye Calibration:

a Python re-implementation and reproduction of Wise et al. (IJRR 2025)

Jerry (Qilong) Cheng

re-implementation, simulation studies and real-data re-run, September 2026

September 5, 2026

Abstract

The robot-world and hand-eye calibration problem $\mathbf{A}_i \mathbf{X} = \mathbf{Y} \mathbf{B}_i$ is non-convex, and local solvers depend on a good initial guess. Wise et al. [1] formulate its generalized version (many sensors $\mathbf{X}_j$, many targets $\mathbf{Y}_k$, monocular sensors with an unknown scale α) as a maximum-likelihood QCQP, eliminate the translations by a Schur complement, and solve the Shor semidefinite relaxation of the remaining rotation problem, which is tight under moderate noise and returns a certificate (the duality gap) with every answer. This document records a from-scratch Python re-implementation (cost assembly, reduction, SDP via SCS/Clarabel through cvxpy, rank-one rounding, a two-stage linear baseline and a local on-manifold solver), the reproduction of the paper’s four simulation studies (100 runs per noise level), and a re-run on the paper’s real-world data (16 AprilTags observed by 8 cameras, 3230 pose measurements, 52+21 camera-tag pairs) that ships with the public repository. The certified solution of the 217-dimensional real problem has a relative duality gap of $-1.09\text{e-}08$ (known scale) and $-1.19\text{e-}08$ (unknown scale), against -6.4×10^{-9} and 8.6×10^{-9} in the paper, and the recovered tag scale $\alpha = 1.0255$ reproduces the paper’s observation that the hand-measured tags were about 2.5% off. In simulation the SDP matches the best-initialised local solver to within a tenth of a millimetre and beats the linear two-stage method, while the same local solver from random starts ends in a local minimum in 26–40% of the runs.

Keywords: hand-eye calibration, robot-world calibration, semidefinite relaxation, certifiable estimation, convex optimization, multi-camera rig.

1 Overview

The published implementation [1] is Julia (JuMP + COSMO/Mosek, utiasSTARS/certifiable-rwhe-calibration). Everything here is a clean Python re-implementation in `experiments/`: `rwhe.py` (problem container, cost matrices, Schur reduction, SDP, rounding, baselines, Langevin sampling, synthetic problems), `sim_studies.py` (paper §8, sharded), `real_data.py` (paper §9 on the repository’s CSV data), `make_tables.py` (tables and the numbers in this text), `make_video.py`. The only pieces taken from the public repository are the real-world measurement files and the noise weights used by its example script ($\kappa = 125$, $\sigma = 0.1$ m).

2 Method

2.1 Problem

For each edge $e = (j, k)$ of a bipartite graph and each measurement i , $\mathbf{A}_{e,i} \mathbf{X}_j = \mathbf{Y}_k \mathbf{B}_{e,i}$ with $\mathbf{A}$ noiseless and $\mathbf{B}$ noisy: $\tilde{\mathbf{t}}_B = \alpha \mathbf{R}_Y^\top (\mathbf{R}_A \mathbf{t}_X + \mathbf{t}_A - \mathbf{t}_Y) + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, and $\tilde{\mathbf{R}}_B = \mathbf{R}_B \mathbf{R}_\epsilon$ with $\mathbf{R}_\epsilon \sim \text{Lang}(\mathbf{I}, \kappa)$. The negative log-likelihood (paper Problem 3) is

$$J = \frac{1}{2} \sum_{e,i} \frac{1}{\sigma_e^2} \|\mathbf{R}_A \mathbf{t}_{X_j} + \alpha \mathbf{t}_A - \mathbf{t}_{Y_k} - \mathbf{R}_{Y_k} \tilde{\mathbf{t}}_B\|^2 + \kappa_e \|\mathbf{R}_A \mathbf{R}_{X_j} - \mathbf{R}_{Y_k} \tilde{\mathbf{R}}_B\|_F^2, \quad (1)$$

where the translation unknowns have absorbed α ($\mathbf{t}_{X,\alpha} = \alpha \mathbf{t}_X$). With $\text{vec}(\mathbf{AXB}) = (\mathbf{B}^\top \otimes \mathbf{A}) \text{vec } \mathbf{X}$ (column-major), every term is a quadratic form in $\mathbf{x} = [\mathbf{t}_{X_1..X_M}, \mathbf{t}_{Y_1..Y_P}, \alpha, \text{vec } \mathbf{R}_{X_1..X_M}, \text{vec } \mathbf{R}_{Y_1..Y_P}]$, so $J = \mathbf{x}^\top \mathbf{Q} \mathbf{x}$ with $\mathbf{Q}$ assembled row by row from $\mathbf{M}_R = [\mathbf{I} \otimes \mathbf{R}_A, -\tilde{\mathbf{R}}_B^\top \otimes \mathbf{I}]$ and $\mathbf{M}_t = [\mathbf{R}_A, -\mathbf{I}, \mathbf{t}_A, -\tilde{\mathbf{t}}_B^\top \otimes \mathbf{I}]$ (paper eqs. 25–29). The known-scale case replaces α by the homogenising variable s with $s^2 = 1$.

2.2 Reduction and relaxation

The translations (and α in the monocular case) enter quadratically and unconstrained, so they are eliminated by the generalized Schur complement $\mathbf{Q}' = \mathbf{Q}_{rr} - \mathbf{Q}_{tr}^\top \mathbf{Q}_{tt}^+ \mathbf{Q}_{tr}$ (paper eq. 48) and recovered afterwards as $[\mathbf{t}; \alpha] = -\mathbf{Q}_{tt}^+ \mathbf{Q}_{tr} [\mathbf{r}; s]$ (eq. 47); the reduced vector is $[\text{vec } \mathbf{R}_1, \dots, \text{vec } \mathbf{R}_n, s]$ with s last (in the known-scale case s is not eliminated because $\mathbf{Q}$ couples it to $\mathbf{t}_A$). Each $\mathbf{R} \in SO(3)$ is encoded by the 21 homogenised quadratic constraints $\mathbf{R} \mathbf{R}^\top = s^2 \mathbf{I}$, $\mathbf{R}^\top \mathbf{R} = s^2 \mathbf{I}$, $\mathbf{c}_p \times \mathbf{c}_q = s \mathbf{c}_r$ (cyclic) (eq. 50), and the Shor relaxation is

$$\min_{\mathbf{Z} \succeq 0} \text{tr}(\mathbf{Q}' \mathbf{Z}) \quad \text{s.t.} \quad Z_{ss} = 1, \quad \text{the 21 linear constraints per rotation on the entries of } \mathbf{Z}, \quad (2)$$

whose optimal value equals the dual Problem 7 of the paper. $\mathbf{Q}'$ is scaled by its largest entry before solving; the solver is SCS at $\epsilon = 10^{-11}$ with a Clarabel fallback (SCS gives $|\hat{\rho}| \sim 10^{-9}$, Clarabel stalls at 10^{-6} on these problems). The solution is rounded by the leading eigenvector of $\mathbf{Z}$ scaled so its s entry is one, projecting each 9-block onto $SO(3)$, and certified by $\hat{\rho} = (p - d)/|d|$ with $p = \mathbf{r}^\top \mathbf{Q}' \mathbf{r}$ at the rounded point and d the SDP value (eqs. 71–73); the eigenvalue ratio λ_2/λ_1 of $\mathbf{Z}$ is reported as a rank check.

2.3 Baselines

Linear two-stage (Shah [2], Wang et al. [3] flavour): smallest eigenvector of the rotation-only cost over $[\text{vec } \mathbf{R}_X, \text{vec } \mathbf{R}_Y]$, blockwise $SO(3)$ projection, translations by linear least squares. *Local solver* (the paper’s LOM stand-in): `scipy.optimize.least_squares` (Levenberg–Marquardt) on the residuals of (1) parameterised by rotation vectors, translations and $\log \alpha$, started from random poses, from the linear solution, or (monocular) from poses 10° and 10 cm from the truth with $\alpha_0 = 1$, exactly the paper’s protocol. Because it minimises the same objective as the SDP, agreement between the two is a direct check of global optimality.

2.4 Noise model

$\text{Lang}(\mathbf{I}, \kappa)$ is sampled by rejection on the angle density $\propto e^{2\kappa \cos \theta} (1 - \cos \theta)$ with a uniform axis; $\kappa = 125$ and 12 correspond to $\approx 3.6^\circ$ and 11.7° per-axis spread.

2.5 Identifiability (paper §6)

The re-implementation checks Corollary 4 on data: an edge is “identifiable alone” if among its $\mathbf{A}$ poses two relative rotations $\mathbf{R}_{A_{i_2}}^\top \mathbf{R}_{A_{i_1}}$, $\mathbf{R}_{A_{i_3}}^\top \mathbf{R}_{A_{i_1}}$ have principal axes more than 5° apart (rotations below 2° are ignored). The graph must be weakly connected for the whole problem to be identifiable (Theorem 3).

3 Experiments

3.1 Robot arm poses on a sphere (paper §8.1, Tables 2–3)

$\mathbf{X} = \mathbf{T}_{hc}$ and $\mathbf{Y} = \mathbf{T}_{bt}$ are random; the camera collects 100 poses on the unit sphere around the target (optical axis toward the target, y toward the south pole; two spheres of radius 1 and 0.3 m in the monocular case, as the paper found necessary). Noise $(\kappa, \sigma) \in \{125, 12\} \times \{1 \text{ cm}, 5 \text{ cm}\}$, $\alpha = 0.5$ when unknown, 100 seeded runs each.

Table 1: Poses on a sphere, known scale: mean $\pm$ std of the translation [mm] and rotation [deg] errors of $\mathbf{X}$ and $\mathbf{Y}$ over 100 runs (paper Table 2).

noise	method	$t_{X,err}$ [mm]	$r_{X,err}$ [deg]	$t_{Y,err}$ [mm]	$r_{Y,err}$ [deg]
$\kappa = 125, \sigma = 1 \text{ cm}$	linear two-stage	7.50 ± 4.88	0.75 ± 0.34	1.92 ± 0.93	0.75 ± 0.34
	local (random init)	362.69 ± 590.64	48.98 ± 79.49	2.06 ± 1.07	48.74 ± 79.79
	local (linear init)	3.58 ± 1.83	0.65 ± 0.24	1.93 ± 0.94	0.22 ± 0.13
	SDP (ours)	3.58 ± 1.83	0.65 ± 0.24	1.93 ± 0.94	0.22 ± 0.13
$\kappa = 125, \sigma = 5 \text{ cm}$	linear two-stage	11.28 ± 5.07	0.74 ± 0.39	9.03 ± 4.24	0.74 ± 0.29
	local (random init)	540.23 ± 648.30	72.26 ± 87.57	9.02 ± 4.33	72.23 ± 87.74
	local (linear init)	10.95 ± 5.31	0.73 ± 0.37	9.03 ± 4.24	0.59 ± 0.32
	SDP (ours)	10.95 ± 5.31	0.73 ± 0.37	9.03 ± 4.24	0.59 ± 0.32
$\kappa = 12, \sigma = 1 \text{ cm}$	linear two-stage	21.70 ± 15.74	2.42 ± 0.99	2.05 ± 0.95	2.26 ± 1.04
	local (random init)	350.07 ± 583.44	47.89 ± 77.53	2.50 ± 1.65	46.97 ± 78.77
	local (linear init)	4.09 ± 2.17	1.97 ± 0.80	2.05 ± 0.93	0.26 ± 0.16
	SDP (ours)	4.09 ± 2.17	1.97 ± 0.80	2.05 ± 0.93	0.26 ± 0.16
$\kappa = 12, \sigma = 5 \text{ cm}$	linear two-stage	23.62 ± 14.32	2.36 ± 1.04	9.04 ± 4.16	2.27 ± 1.03
	local (random init)	385.60 ± 594.33	51.38 ± 79.29	9.11 ± 4.13	50.97 ± 80.23
	local (linear init)	15.85 ± 7.07	2.08 ± 0.93	9.05 ± 4.14	0.95 ± 0.49
	SDP (ours)	15.85 ± 7.07	2.08 ± 0.93	9.05 ± 4.14	0.95 ± 0.49

Table 2: Poses on two spheres, unknown scale (paper Table 3).

noise	method	$t_{X,err}$ [mm]	$r_{X,err}$ [deg]	$t_{Y,err}$ [mm]	$r_{Y,err}$ [deg]	α_{err} [%]
$\kappa = 125, \sigma = 1 \text{ cm}$	local (close init)	5.99 ± 2.71	0.64 ± 0.26	3.96 ± 1.70	0.35 ± 0.16	$4.4\text{e-}01 \pm 3.8\text{e-}01$
	SDP (ours)	5.99 ± 2.71	0.64 ± 0.26	3.96 ± 1.70	0.35 ± 0.16	$4.4\text{e-}01 \pm 3.8\text{e-}01$
$\kappa = 125, \sigma = 5 \text{ cm}$	local (close init)	26.33 ± 11.86	0.68 ± 0.27	19.62 ± 8.43	0.67 ± 0.28	$2.2\text{e+}00 \pm 1.6\text{e+}00$
	SDP (ours)	26.33 ± 11.86	0.68 ± 0.27	19.62 ± 8.43	0.67 ± 0.28	$2.2\text{e+}00 \pm 1.6\text{e+}00$
$\kappa = 12, \sigma = 1 \text{ cm}$	local (close init)	5.54 ± 2.56	1.83 ± 0.75	3.75 ± 1.63	0.37 ± 0.17	$5.1\text{e-}01 \pm 3.7\text{e-}01$
	SDP (ours)	5.54 ± 2.56	1.83 ± 0.75	3.75 ± 1.63	0.37 ± 0.17	$5.1\text{e-}01 \pm 3.7\text{e-}01$
$\kappa = 12, \sigma = 5 \text{ cm}$	local (close init)	29.42 ± 12.46	2.10 ± 0.96	19.43 ± 7.99	1.58 ± 0.77	$2.6\text{e+}00 \pm 1.6\text{e+}00$
	SDP (ours)	29.42 ± 12.46	2.10 ± 0.96	19.43 ± 7.99	1.58 ± 0.77	$2.6\text{e+}00 \pm 1.6\text{e+}00$

Tables 1–2 and Fig. 1 reproduce the paper’s ordering: the certified SDP and the well-initialised local solver agree to within 0.1 mm and 0.01° (they minimise the same objective and the SDP is tight), both beat the linear two-stage method (3.6 vs 7.5 mm at $\kappa = 125, \sigma = 1$ cm; 15.9 vs 23.6 mm at $\kappa = 12, \sigma = 5$ cm), and the local solver from random starts converges to a local minimum in 26–40 % of the runs with mean errors of hundreds of millimetres (363 mm at the lowest noise), which is why the paper omits it from its tables. With the scale free, the random-start local solver diverges outright (the estimated α runs away and the translations with it), so it is left out of Tables 2 and 4. The absolute numbers differ from the paper’s Table 2 (which lists 10.9 mm/ 0.77° for the same setting) because the sphere geometry and the random $\mathbf{X}, \mathbf{Y}$ are not specified there; the ratios between methods are what is reproduced. In the monocular case the scale is recovered to 4.4e-01 % ($\sigma = 1$ cm) and 2.2e+00 % ($\sigma = 5$ cm), the paper’s 10^{-3} – 10^{-2} % range.

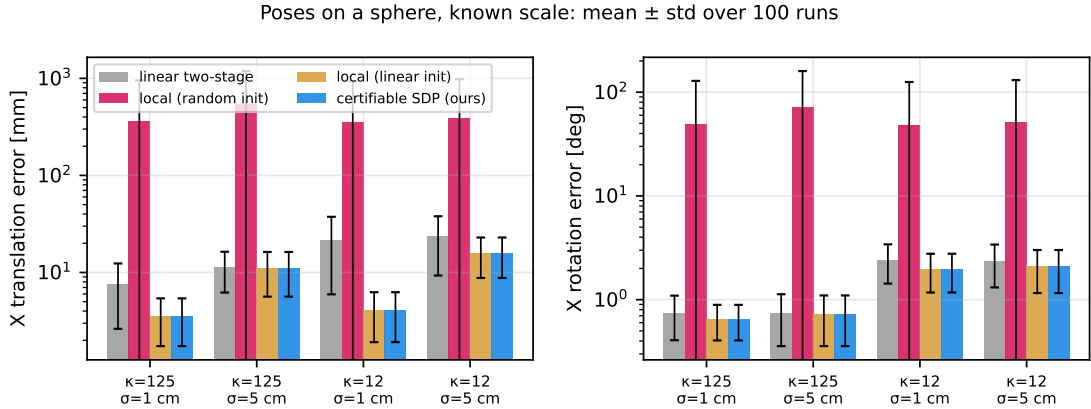


Figure 1: Poses on a sphere, known scale: $\mathbf{X}$ errors per method and noise level.

3.2 Four cameras observing a hand-mounted target (paper §8.2, Tables 4–5)

Four fixed cameras at about 1.5 m observe a target on the hand of an arm that visits 108 poses on two spheres (0.4/0.15 m); in the $\mathbf{AX} = \mathbf{YB}$ convention the shared hand–target transform is the single $\mathbf{X}$ and the four camera poses are the $\mathbf{Y}_k$ (one edge per camera, five rotations, 46×46 SDP). Tables 3–4 show the same pattern as the paper’s Tables 4–5: the SDP and the local solver agree, the camera rotations are recovered to 0.31 versus 0.58 degrees for the linear method at the lowest noise, and the shared transform to 1.1 mm. The camera translation errors (7.9 mm) are larger than the paper’s because a 1° rotation error at a 1.5 m lever arm is already 26 mm; the paper does not state its camera distances.

3.3 Certificate

Table 5 and Fig. 2 list the duality gap, rank ratio and the local-minimum rate of the random-start local solver for every study: the largest $|\hat{\rho}|$ over all 16×100 solves is 1.9e-08 and the largest λ_2/λ_1 is 3.4e-11, i.e. the relaxation is tight in every run, including $\kappa = 12, \sigma = 5$ cm (the paper’s Proposition 7 only promises tightness for small enough noise). A single-pair SDP solves in at most 22.23 s.

Table 3: Four cameras, known scale (paper Table 4); t_Y, r_Y are averaged over the four cameras.

noise	method	$t_{X,err}$ [mm]	$r_{X,err}$ [deg]	$t_{Y,err}$ [mm]	$r_{Y,err}$ [deg]
$\kappa = 125, \sigma = 1 \text{ cm}$	linear two-stage	1.34 ± 0.67	0.36 ± 0.12	14.36 ± 4.17	0.58 ± 0.13
	local (random init)	42.40 ± 132.96	16.48 ± 51.37	234.72 ± 725.11	16.42 ± 51.22
	local (linear init)	1.13 ± 0.50	0.33 ± 0.11	7.87 ± 2.67	0.31 ± 0.09
	SDP (ours)	1.13 ± 0.50	0.33 ± 0.11	7.87 ± 2.67	0.31 ± 0.09
$\kappa = 125, \sigma = 5 \text{ cm}$	linear two-stage	4.70 ± 2.21	0.37 ± 0.15	17.00 ± 4.43	0.60 ± 0.14
	local (random init)	119.99 ± 196.30	52.40 ± 81.41	775.58 ± 1193.33	52.49 ± 81.26
	local (linear init)	4.67 ± 2.22	0.37 ± 0.15	16.24 ± 4.35	0.56 ± 0.13
	SDP (ours)	4.67 ± 2.22	0.37 ± 0.15	16.24 ± 4.35	0.56 ± 0.13
$\kappa = 12, \sigma = 1 \text{ cm}$	linear two-stage	3.02 ± 2.23	1.07 ± 0.52	48.59 ± 13.96	1.92 ± 0.44
	local (random init)	6.16 ± 50.13	2.67 ± 17.82	37.96 ± 281.75	2.18 ± 17.85
	local (linear init)	1.13 ± 0.54	0.88 ± 0.37	9.63 ± 3.68	0.38 ± 0.13
	SDP (ours)	1.13 ± 0.54	0.88 ± 0.37	9.63 ± 3.68	0.38 ± 0.13
$\kappa = 12, \sigma = 5 \text{ cm}$	linear two-stage	5.61 ± 2.43	1.11 ± 0.46	47.16 ± 13.70	1.99 ± 0.45
	local (random init)	62.69 ± 152.15	25.97 ± 61.85	398.15 ± 912.46	26.11 ± 61.44
	local (linear init)	5.08 ± 2.06	1.01 ± 0.44	32.22 ± 10.13	1.33 ± 0.36
	SDP (ours)	5.08 ± 2.06	1.01 ± 0.44	32.22 ± 10.13	1.33 ± 0.36

Table 4: Four cameras, unknown scale (paper Table 5).

noise	method	$t_{X,err}$ [mm]	$r_{X,err}$ [deg]	$t_{Y,err}$ [mm]	$r_{Y,err}$ [deg]	α_{err} [%]
$\kappa = 125, \sigma = 1 \text{ cm}$	local (close init)	2.99 ± 1.68	0.31 ± 0.15	17.17 ± 6.21	0.45 ± 0.11	$6.1\text{e-}01 \pm 4.5\text{e-}01$
	SDP (ours)	2.99 ± 1.68	0.31 ± 0.15	17.17 ± 6.21	0.45 ± 0.11	$6.1\text{e-}01 \pm 4.5\text{e-}01$
$\kappa = 125, \sigma = 5 \text{ cm}$	local (close init)	14.57 ± 8.52	0.35 ± 0.16	57.41 ± 33.30	0.59 ± 0.12	$2.9\text{e+}00 \pm 2.1\text{e+}00$
	SDP (ours)	14.57 ± 8.52	0.35 ± 0.16	57.41 ± 33.30	0.59 ± 0.12	$2.9\text{e+}00 \pm 2.1\text{e+}00$
$\kappa = 12, \sigma = 1 \text{ cm}$	local (close init)	3.18 ± 1.40	0.92 ± 0.38	21.86 ± 7.97	0.69 ± 0.24	$5.9\text{e-}01 \pm 4.4\text{e-}01$
	SDP (ours)	3.18 ± 1.40	0.92 ± 0.38	21.86 ± 7.97	0.69 ± 0.24	$5.9\text{e-}01 \pm 4.4\text{e-}01$
$\kappa = 12, \sigma = 5 \text{ cm}$	local (close init)	15.05 ± 7.26	1.01 ± 0.46	75.29 ± 31.46	1.73 ± 0.38	$3.1\text{e+}00 \pm 2.4\text{e+}00$
	SDP (ours)	15.05 ± 7.26	1.01 ± 0.46	75.29 ± 31.46	1.73 ± 0.38	$3.1\text{e+}00 \pm 2.4\text{e+}00$

3.4 Real-world data (paper §9)

The repository ships the processed measurements of the paper’s experiment: a mobile rig with eight hardware-synchronised Point Grey Blackfly cameras tracked by OptiTrack ($\mathbf{A}_{(j,k),i}$) observing 16 AprilTags in a motion-capture room ($\mathbf{B}_{(j,k),i}$ from the tag detector), 3230 measurements over 73 camera–tag pairs of which 52 satisfy Corollary 4 on their own and 21 need the rest of the graph (Fig. 3; the paper’s Fig. 10 shows the same blue/red pattern). With $\kappa = 125$ and $\sigma = 0.1 \text{ m}$ the joint problem has 24 rotations, i.e. a 217-dimensional PSD variable with 24×21 constraints; SCS solves it in 230 s (known scale) and 81 s (unknown scale) on a laptop, the paper’s COSMO solve took “approximately five minutes”.

Certificate and scale. $\hat{\rho} = -1.09\text{e} - 08$ (known scale) and $-1.19\text{e-}08$ (unknown scale), with $\lambda_2/\lambda_1 = 7.7\text{e} - 13$: the same 10^{-8} – 10^{-9} floating-point floor as the paper’s -6.41×10^{-9} and 8.55×10^{-9} . The monocular solve returns $\alpha = 1.0255$ on the tag-derived translations, i.e. the AprilTags were 2.5% smaller than the size used to compute $\mathbf{B}$, exactly the paper’s finding (“the estimated AprilTag scale is 2.5% smaller than the hand-measured value”); the median loop-closure residual $|\mathbf{AX} - \mathbf{YB}|$ is 2.52 cm and 1.30 degrees. The SDP cost (985.244) equals the local solver’s from the linear initialisation (985.244), while the linear two-stage solution is at 1027.278 and differs

Table 5: Certificate statistics per study and noise level.

study	κ	σ [cm]	median $\hat{\rho}$	max $ \hat{\rho} $	max λ_2/λ_1	local minima (random init)	SDP time [s]
sphere, known scale	125	1	-8.9e-10	1.7e-08	3.8e-12	27%	0.39
sphere, known scale	125	5	-3.1e-09	1.2e-08	2.2e-11	40%	3.44
sphere, known scale	12	1	+1.1e-12	1.9e-08	6.0e-12	26%	0.26
sphere, known scale	12	5	+9.3e-11	4.1e-10	4.0e-12	28%	0.24
sphere, unknown scale	125	1	-6.2e-10	5.0e-09	8.7e-12	28%	2.36
sphere, unknown scale	125	5	-4.4e-09	1.9e-08	3.4e-11	33%	3.51
sphere, unknown scale	12	1	-5.1e-13	2.6e-09	4.2e-12	24%	0.25
sphere, unknown scale	12	5	-2.0e-10	9.0e-10	1.9e-11	33%	3.44
4 cameras, known scale	125	1	+1.3e-09	7.5e-09	1.0e-11	9%	16.00
4 cameras, known scale	125	5	+1.4e-09	5.2e-09	6.7e-12	29%	11.97
4 cameras, known scale	12	1	+1.3e-10	7.4e-09	9.6e-12	1%	4.82
4 cameras, known scale	12	5	+1.2e-10	5.7e-10	8.5e-12	14%	8.36
4 cameras, unknown scale	125	1	+9.7e-10	6.0e-09	8.3e-12	24%	8.32
4 cameras, unknown scale	125	5	+1.0e-09	4.9e-09	6.3e-12	25%	6.85
4 cameras, unknown scale	12	1	+1.0e-10	3.4e-09	8.8e-12	17%	4.56
4 cameras, unknown scale	12	5	+1.3e-10	5.4e-10	7.5e-12	31%	5.69

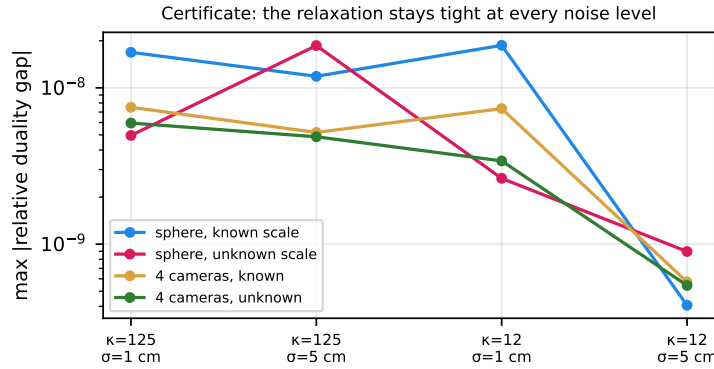


Figure 2: Maximum relative duality gap per study and noise level.

from the certified one by up to 3.78 cm and 1.12 degrees per camera.

Camera rig. Table 6 lists the eight camera poses relative to camera 0 (Fig. 4). The Kalibr ground truth of the paper is not part of the repository, so the paper’s Table 6 (3.2 cm/1.0° mean error against Kalibr) cannot be recomputed; what can be checked is internal: the certified solution, the local solver and the unknown-scale solution agree on the camera geometry to within a few millimetres, and the baselines (22–81 cm) are consistent with a hand-sized multi-camera rig.

4 Implementation and Usage

```
python3 -m venv .venv
.venv/bin/pip install numpy scipy matplotlib cvxpy scs clarabel
cd experiments
for k in 0 1 2 3 4; do
    python sim_studies.py 100 --shard $k --nshards 5 &
done; wait
```

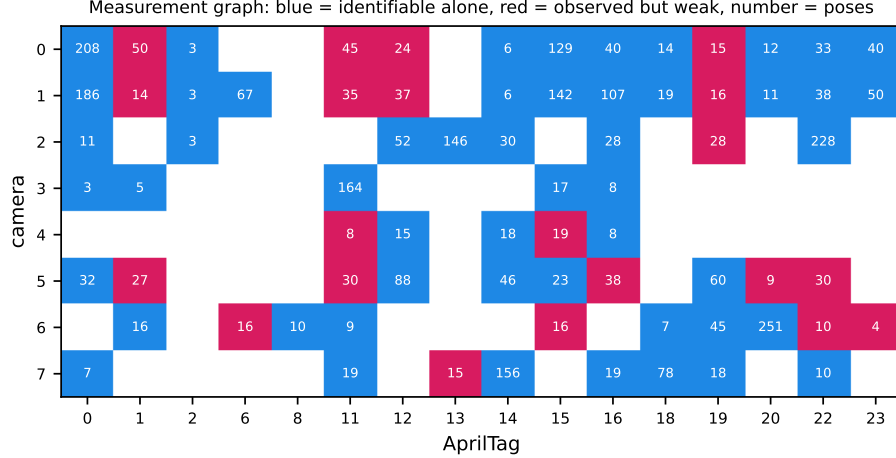


Figure 3: Which camera saw which tag: blue = identifiable alone, red = observed but insufficient excitation on its own, numbers = poses per pair.

Table 6: Camera poses in the frame of camera 0 from the certified known-scale solution, the baseline from the unknown-scale solution, and the distance to the linear two-stage solution.

camera	baseline to cam 0 [cm]	roll [deg]	pitch [deg]	yaw [deg]	baseline, unknown scale [cm]	$ \Delta t $ vs linear [cm]	$ \Delta R $ vs linear [deg]
0	0.0	0.0	0.0	0.0	0.0	1.94	0.99
1	22.2	-2.1	-2.8	-0.9	21.5	2.66	1.05
2	25.0	-29.2	-56.9	0.4	25.6	1.03	0.89
3	43.9	17.1	47.0	-1.6	44.4	3.78	1.12
4	41.3	44.0	-81.8	-32.0	42.4	2.15	0.76
5	64.1	118.3	-83.8	-122.8	65.2	0.44	0.86
6	80.6	173.8	78.4	172.7	81.2	1.04	0.91
7	57.6	-120.0	79.3	-87.0	59.0	0.46	0.76

```
python sim_studies.py          # merge shards -> out/sim_studies.json + figures
python real_data.py           # real rig: out/real_data.json + figures (~5 min)
python make_tables.py && (cd ../docs && make)
python make_video.py
```

`Problem(M, P, edges, scale_free)` holds the graph; `sdp_solve` returns the poses, α , d , p , $\hat{p}$; `linear_solve` and `local_solve` are the baselines; `sphere_problem` and `multicam_problem` generate the studies. Hardware path: any sensor that returns a pose of a target (fiducial detector, ICP, or a scaled monocular pose) provides **B**, and any pose source for the platform (encoders, motion capture) provides **A**.

5 Limitations

- The synthetic geometries (sphere radius, camera placement, random **X**, **Y**) are re-created from the paper’s description, so absolute error magnitudes are not comparable with its tables; method rankings and certificate behaviour are.
- The paper’s other baselines (Dornaika & Horaud, Horn et al. dual-quaternion SDP) were not re-implemented.
- The local solver uses chordal rotation residuals (the SDP objective) rather than the paper’s LOM exponential-map noise model; this makes the SDP/local comparison exact but is not

8 cameras in the frame of camera 0 (red = optical axes)

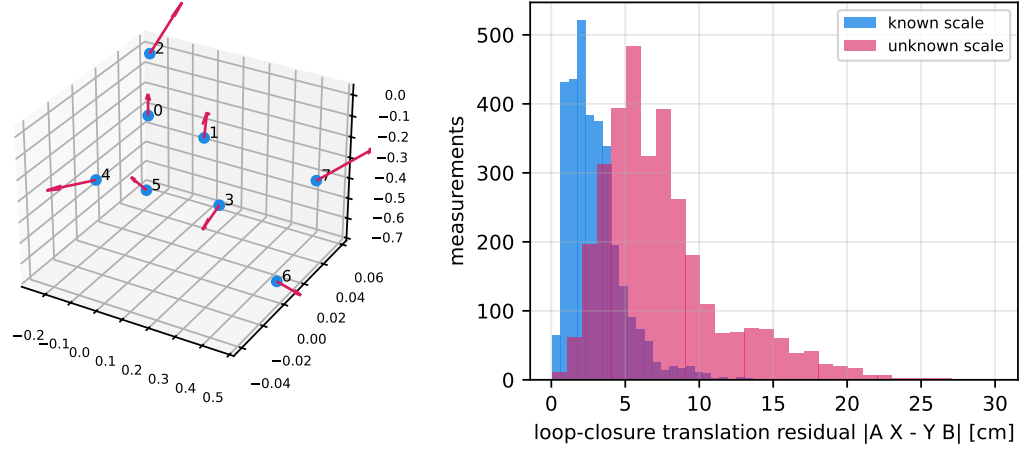


Figure 4: The eight cameras (optical axes in red) and the loop-closure residual distribution.

the paper’s LOM.

- cvxpy’s dense PSD variable limits the size: the 217-dimensional real problem takes minutes, and a problem with hundreds of cameras would need the sparse chordal structure the paper exploits with COSMO.
- No Kalibr or OptiTrack ground truth for the real rig is available, so accuracy on the real data is reported as internal consistency only.

References

- [1] E. Wise, P. Kaveti, Q. Cheng, W. Wang, H. Singh, J. Kelly, D. M. Rosen, M. Giamou, “A Certifiably Correct Algorithm for Generalized Robot-World and Hand-Eye Calibration,” *Int. J. Robotics Research*, 2025. arXiv:2507.23045.
- [2] M. Shah, “Solving the robot-world/hand-eye calibration problem using the Kronecker product,” *J. Mechanisms and Robotics*, 2013.
- [3] G. Wang et al., “Accurate and efficient hand-eye and robot-world calibration with multiple cameras/targets,” 2022.
- [4] D. M. Rosen, L. Carlone, A. S. Bandeira, J. J. Leonard, “SE-Sync: a certifiably correct algorithm for synchronization over the special Euclidean group,” *IJRR*, 2019.
- [5] D. Cifuentes, S. Agarwal, P. A. Parrilo, R. R. Thomas, “On the local stability of semidefinite relaxations,” *Math. Programming*, 2022.
- [6] J. Heller, D. Henrion, T. Pajdla, “Hand-eye and robot-world calibration by global polynomial optimization,” *ICRA*, 2014.
- [7] R. Tron, D. M. Rosen, L. Carlone, “On the inclusion of determinant constraints in Lagrangian duality for 3D SLAM,” *RSS workshop*, 2015.